

RESEARCH POSITIONING PAPER

Pramāṇa-Constrained Reasoning

Nyāya Darshan as a Computational Foundation for Explainable and Responsible AI

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Contemporary explainable AI (XAI) treats explanation as a post-hoc artefact. A model produces a decision, and a separate mechanism then constructs a plausible account of why that decision occurred. This architecture creates a structural faithfulness problem: the explanation is generated *about* the reasoning rather than *by* it, and no formal guarantee connects the two.

This paper argues that Nyāya Darshan, the classical Indian school of logic and epistemology, offers a computationally tractable alternative in which justification is a precondition of inference rather than a commentary upon it. Nyāya’s three central constructs (pramāṇa, the classified means of valid knowledge; hetvābhāsa, the taxonomy of defective reasons; and pañcāvayava, the five-membered demonstrative syllogism) map with unusual directness onto the requirements that emerging AI regulation now imposes on automated decision systems.

The core claim is architectural, not merely cultural: a system that must satisfy source-quality conditions *before* it is permitted to infer is explainable by construction, produces a natively contestable output, and can refuse to reason when its epistemic conditions are unmet. A working reference implementation of these ideas was built and validated in early 2026 as part of Unriddling Inference 2026 at IIT Delhi. This paper sets out the intellectual case, the market gap, the open research questions, and the honest limits of the approach.

1. The Problem: Explanation as Decoration

1.1 What the black box actually is

The phrase “black box” is often used loosely to mean “complicated.” The precise problem is narrower and more serious. In a deep network, the decision is distributed across millions of parameters in a form that admits no compact, human-legible description. There is no location in the model where the reason lives. The decision is real; the reason, in any sense a human would recognise, does not exist as a retrievable object.

This is why the dominant XAI methods are *attribution* methods rather than *justification* methods. SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) do not recover a reason. They estimate how much each input feature moved the output. That is a genuinely valuable measurement, and this paper does not dispute its rigour. Shapley values carry real axiomatic guarantees (efficiency, symmetry, dummy, additivity) that few alternatives can match.

But a feature attribution answers a different question from the one a human asks.

1.2 The question mismatch

When a clinician, a loan applicant, a patient, or a regulator asks “why,” they are not asking for a gradient. They are asking a compound question with at least four parts:

1. **What is the claim?**
2. **On what grounds does it rest?**
3. **How reliable are those grounds, and how were they obtained?**
4. **Where could this reasoning fail, and how would I contest it?**

A SHAP plot answers none of these. It answers “which inputs mattered most,” which is an engineering diagnostic, not a justification. The mismatch is not a presentation problem to be solved with better charts. It is a category difference between *attribution* and *warrant*.

1.3 The faithfulness gap

Because post-hoc explanation is generated separately from the decision, it can be plausible without being true. This is well documented in the literature: explanations can be manipulated adversarially, can vary substantially with sampling choices, and can present a coherent narrative for a decision that was in fact driven by a spurious correlate. The explanation is optimised for human acceptability, not for correspondence with the underlying computation.

An architecture in which the explanation is generated after the decision can never fully close this gap. The gap is a consequence of the ordering.

2. The Market Gap

2.1 What exists

The XAI toolchain is mature along one axis and empty along another.

Well served: feature attribution (SHAP, LIME, Integrated Gradients), counterfactual generation, saliency mapping, concept activation vectors, surrogate modelling.

Poorly served or absent:

Capability	Status in current XAI
Evidence quality assessment (is this source trustworthy, and how do we know?)	Absent. Inputs are treated as epistemically uniform.
Reasoning defect detection (is the inference itself malformed?)	Absent. No standard tool checks the <i>form</i> of the inference.
Structured contestability (can a user challenge a specific step?)	Absent. A user can reject an output but cannot dispute a link in the chain.
Principled abstention (should the system answer at all?)	Weak. Confidence thresholds exist, but confidence is not the same as epistemic sufficiency.
Temporal validity of evidence (is this input still true?)	Absent. Staleness is a data engineering concern, not a reasoning concern.

2.2 The regulatory pressure that widens the gap

The EU AI Act, in Article 86, establishes a right to explanation for individual decision-making in high-risk contexts. It requires “clear and meaningful explanations of the role of the AI system in the decision-making procedure and the main elements of the decision taken.” The NIST AI Risk Management Framework asks for systems that are “explainable and interpretable” alongside “valid and reliable.” ISO/IEC 42001 requires documented, auditable AI management processes.

None of these frameworks are satisfied by a feature importance ranking. A regulator asking “on what basis was this person denied” is not asking for coefficients. The compliance instruments are converging on a demand for *warrant*, and the tooling ecosystem supplies *attribution*.

This is the gap. It is not a niche. It is the difference between what regulation is about to require and what the field currently produces.

2.3 The cultural gap, stated precisely

There is a second, less discussed gap. Contemporary XAI is built almost entirely on a Western analytic scaffolding: propositional logic, probability theory, statistical decision theory. This is not a criticism of its rigour. It is an observation about coverage.

Domains grounded in non-Western epistemic traditions (Ayurveda, classical Indian jurisprudence, traditional agricultural and ecological knowledge systems) are not merely underserved by these tools. They are structurally mismatched. An Ayurvedic recommendation is not the output of a statistical model over population data. It is the output of a reasoning tradition with its own criteria for what counts as valid grounds, its own conception of individual constitution, and its own theory of when a general rule fails to apply to a particular case.

Wrapping such a system in SHAP does not explain it. It translates it into a vocabulary that cannot express its actual logic.

3. Why Nyāya Darshan

3.1 What Nyāya is, and why it is unusually suitable

Nyāya is one of the six orthodox (āstika) darshanās of Indian philosophy, systematised in Gautama’s *Nyāya Sūtra* (c. 2nd century BCE to 2nd century CE) and elaborated across two millennia through Vātsyāyana’s *Nyāya Bhāṣya*, Uddyotakara, Vācaspati Miśra, Udayana, and the Navya Nyāya tradition beginning with Gaṅgeśa.

Its distinguishing feature, relative to most Western logical traditions, is that it is *primarily an epistemology and only secondarily a logic*. The central question is not “what follows from what” but “how do you know, and what kind of knowing is it.” Nyāya begins from the question that XAI has been unable to answer.

Three properties make it computationally attractive:

It is explicitly source-typed. Nyāya does not treat knowledge as a homogeneous substance. It classifies knowledge by acquisition route, and it treats those routes as having different epistemic weight. This is precisely the missing capability identified in Section 2.1.

It has a defect taxonomy. Nyāya does not merely define valid inference. It enumerates the specific ways inference goes wrong, as named, recognisable categories. This is a ready-made specification for a reasoning validator.

Its canonical output form is a demonstration, not an assertion. The pañcāvayava (five-membered) syllogism is not a compressed proof for logicians. It is a demonstrative form designed for a public disputation setting, intended to make each step separately inspectable and separately challengeable by an opponent. It is, in modern terms, an audit trail with contestability built into its grammar.

3.2 Pramāṇa: the source-typing mechanism

Nyāya recognises four pramāṇa (means of valid knowledge). Each answers “how was this known?” differently:

Pramāṇa	Meaning	Epistemic character	Computational analogue
Pratyaksha	Direct perception	Immediate, unmediated cognition	Sensor reading, direct measurement, first-party self-report
Anumāna	Inference from a mark	Derived via an established invariable relation	Model output, derived feature, logical deduction from observed signs
Śabda	Reliable testimony	Received from a trustworthy source	Literature, expert statement, canonical reference, documented guideline
Upamāna	Comparison or analogy	Established by structural similarity to a known case	Case-based reasoning, cohort matching, nearest-neighbour precedent

Nyāya holds these in a strength ordering. Where direct perception conflicts with testimony, perception prevails. Vātsyāyana’s commentary is explicit that pratyaksha grounds the others: inference depends on a previously perceived invariable relation, and testimony depends on the reliability of a perceiver.

Rendered computationally, this yields an ordering (Pratyaksha > Anumāna > Śabda > Upamāna) that is not an arbitrary weighting scheme. It is a defensible epistemological commitment with a textual basis, which matters enormously for auditability. A system that must justify *why* it weighted a lab result above a clinical guideline can cite a position, not a hyperparameter.

This is the first thing current XAI cannot do. SHAP has no mechanism to distinguish a feature derived from a direct measurement from one derived from a questionnaire response from one imputed from a population average. All three enter as numbers. Pramāṇa typing makes the distinction first-class.

3.3 Hetvābhāsa: the defect taxonomy

Nyāya enumerates hetvābhāsa, the “fallacious reasons”: inferences that have the surface form of a valid argument but fail on inspection. The classical five are savyabhicāra (the deviant or straying reason), viruddha (the contradictory reason), satpratipakṣa (the counterbalanced reason), asiddha (the unestablished reason), and bādhita (the reason defeated by stronger evidence).

The most operationally significant of these for AI is **savyabhicāra**, the irregular reason. A reason is savyabhicāra when it is present in cases where the conclusion holds *and* in cases where it does not. In modern terms, this is overgeneralisation: a rule true in aggregate that fails in the specific instance.

This is the dominant failure mode of statistical recommendation systems. A model trained on population data learns that intervention X correlates with outcome Y, and then applies X to an individual for whom the correlation inverts. There is no standard XAI mechanism that detects this. SHAP will faithfully report that the model relied on the feature encoding X, which is true and useless. Hetvābhāsa detection asks a different question: *does the reason offered actually license this conclusion in this case, or does it also license the opposite?*

This is the second thing current XAI cannot do. There is no SHAP equivalent to a fallacy detector, because attribution methods explain what the model did, not whether what it did was warranted.

3.4 Pañcāvayava: the demonstrative output form

The five members are:

1. **Pratijñā** (thesis): the claim to be established.
2. **Hetu** (reason): the ground on which it rests.
3. **Udāharaṇa** (exemplification): the general rule (vyāpti, invariable concomitance) together with a supporting instance.
4. **Upanaya** (application): the explicit application of the general rule to the present case.
5. **Nigamana** (conclusion): the thesis restated as established.

Western logic, since Aristotle, has generally regarded steps 3 and 4 as redundant. Nyāya retains them deliberately, because the tradition is oriented toward *public demonstration* rather than *formal minimality*. Step 3 forces the arguer to state the general principle explicitly, exposing it to challenge. Step 4 forces explicit application, exposing the claim that this case actually falls under the principle.

The redundancy that a formal logician removes is exactly the redundancy that an auditor requires.

Mapped to regulation, the five members correspond closely to what Article 86 asks for: the decision (pratijñā), the main elements it rested on (hetu), the rule applied (udāharaṇa), the role of the system in applying that rule to this individual (upanaya), and the outcome (nigamana). This correspondence is the strongest single argument for the framework’s practical relevance, and it deserves formal study rather than assertion. It is listed in Section 7 as an open research question, not a settled result.

4. The Novel Contribution: Inverting the Order of Explanation

4.1 The central architectural claim

The proposal is a single inversion.

Current XAI: decide, then explain. **Pramāṇa-constrained reasoning:** justify, then decide.

In the proposed architecture, an inference is not permitted to complete unless it satisfies epistemic preconditions. Concretely:

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observe() → incoming evidence is typed by pramāṇa,
           stored with source, confidence, and timestamp

infer() → candidate conclusion is assembled from held beliefs

validate() → PRAMĀṆA GATE: is source coverage sufficient?
             (minimum diversity of source types; at least one
             strong source; no reliance on analogy alone)
             HETVĀBHĀSA SCAN: is the reason itself defective?
             (does it stray, contradict, beg the question,
             rest on an unestablished premise, or rely on
             evidence that has expired?)

explain() → a passing inference emits its pañcāvayava trace

refuse() → a failing inference emits its failure reason

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The explanation is not produced by a separate module. It is the record of the process that produced the decision. Faithfulness is not achieved through better approximation. It is achieved by construction, because there is nothing else the trace could be a record of.

This is the difference between **explainable by design** and **explainable by afterthought**, and it is the paper’s principal claim.

4.2 Refusal as a first-class output

A consequence of the inversion is that the system acquires a capability most models lack: the ability to decline.

Standard classifiers always emit a class. Standard recommenders always emit a recommendation. Confidence scores exist, but a low-confidence answer is still an answer, and the threshold at which to withhold it is a deployment decision external to the model’s reasoning.

Under pramāṇa constraint, abstention is internal and principled. If the available evidence lacks source diversity, the system does not produce a low-confidence recommendation. It produces a refusal that names what is missing. This distinction matters in practice: “I am 54% confident” is a number a user must interpret; “I cannot advise, because my only current information about your state is ten days old” is an actionable statement that tells the user exactly what to do next.

4.3 Temporal validity as an epistemic property

Classical Nyāya includes bādhita, the reason defeated by stronger contrary evidence. Practical AI deployment surfaces a related but distinct condition: evidence that was valid when obtained and has since expired.

The reference implementation treats this as a named condition (kālatīta, the untimely reason) and applies it asymmetrically. Perceptual evidence about a changing state has a freshness requirement; textual authority does not expire in the same way. A user’s self-report from ten days ago is no longer a reliable statement about their present condition. A citation from a classical text is not degraded by the passage of ten days.

A note on intellectual honesty: *kālātīta* as deployed here is an extension rather than a canonical *hetvābhāsa*. It is grounded in *bādhita* and in the tradition’s general sensitivity to the conditions under which a reason holds, but it is not a straightforward reproduction of a classical category. This is flagged explicitly in the implementation and should be flagged in any publication. Extensions to a living tradition are legitimate; presenting them as canonical is not.

Notably, an expired-evidence condition should produce *suspension* rather than *rejection*. Stale data does not mean the reasoning was malformed. It means the reasoning cannot currently be grounded. The distinction between “your logic is wrong” and “your information is out of date” is philosophically real and operationally useful, and it is invisible to any system that models validity as a single scalar.

5. Positioning Relative to Existing Methods

5.1 The relationship to SHAP and LIME is complementary, not competitive

This must be stated carefully, because overclaiming here is the fastest way to lose a technical audience.

SHAP is a quantitative attribution method with formal axiomatic guarantees. The framework proposed here is a qualitative justification method with no comparable formal guarantees at present. They answer different questions and operate at different levels.

Dimension	SHAP / LIME	Pramāṇa-constrained reasoning
Question answered	How much did each feature contribute?	On what grounds should this be believed?
Output form	Numeric attribution vector	Human-legible five-step demonstration
Formal guarantees	Strong (Shapley axioms)	None yet established
Evidence quality modelling	None (inputs are uniform)	Source-typed with a defensible ordering
Reasoning defect detection	None	Five named defect categories
Contestability	User may reject the output	User may challenge a specific member of the chain
Abstention	External thresholding	Internal, principled, with a stated cause
Computational cost	Moderate to high	Low if template-driven; high if generative
Cultural coverage	Statistical / Western analytic	Native to Indian knowledge systems; generalises to any domain with articulable warrant criteria

The productive framing is a two-layer stack. Attribution methods supply a quantitative substrate that constrains what a justification layer is permitted to assert. A justification narrative that

contradicts the attribution evidence should be blocked. Used this way, SHAP becomes a faithfulness check on the reasoning layer, and the reasoning layer becomes the human-facing surface that attribution cannot provide.

Positioning this work as a *replacement* for SHAP is both false and strategically unwise. Positioning it as the layer that sits above attribution, consuming it and producing warrant, is defensible.

5.2 Relationship to neuro-symbolic AI and formal argumentation

The proposal is not without neighbours. Neuro-symbolic architectures pursue similar goals of combining learned representation with symbolic structure. Formal argumentation frameworks (Dung’s abstract argumentation, ASPIC+, defeasible logic) model attack, defeat, and reinstatement with considerable mathematical sophistication. Provenance research in databases (W3C PROV) formalises source tracking.

An honest positioning acknowledges these and states the differentiator precisely. The claim is not that source tracking is novel, nor that defeasible reasoning is novel. The claim is that **Nyāya supplies an integrated package**: a source taxonomy with a principled ordering, a defect taxonomy, and a demonstrative output form, all developed together within a single tradition over two millennia of adversarial refinement in actual disputation. The components have analogues; the coherent integration, and its native fit to certain knowledge domains, is what is offered.

A reviewer at FAccT, AAIL, or IJCAI will ask exactly this question. The answer should be ready before the question is.

6. Reference Implementation and Validation

A working implementation was constructed and tested in March 2026, and presented at Unriddling Inference 2026, a hackathon and conference on computational inference hosted at IIT Delhi and organised by the Institute for Science and Spirituality (Delhi), the IKS Division of the Ministry of Education, and Bṛhat. The judging panel combined philosophers and computer scientists, which imposed a useful dual constraint: the philosophy had to be genuine, and the system had to run.

The implementation is deliberately logic-first and model-free. It contains no machine learning components, which isolates the epistemic architecture as the object of study rather than confounding it with model behaviour.

Five scenarios were constructed to exercise distinct epistemic behaviours, three in a wellness domain (where Ayurvedic constitutional reasoning provides a natural test bed) and two in general domains (to demonstrate that the framework is not domain-bound):

Scenario	Condition tested	Pramāṇa-constrained agent	Source-agnostic baseline
Full source coverage, no defect	Normal acceptance	Accepted, with complete demonstrative trace	Accepted, no trace
Perceptual evidence 12 days stale	Temporal validity	Suspended, requested current information	Accepted, using expired data
Warming intervention for a constitutionally heat-dominant individual with existing inflammatory symptoms	Savyabhicāra (overgeneralisation)	Rejected, identified the straying reason	Accepted, would have caused harm
Direct measurement contradicting a documented guideline	Source-strength revision	Belief revised, with the ordering cited as the reason	Silently overwritten, no record
Documented study contradicting recent derived analysis	Conflict handling	Surfaced the conflict, declined false confidence	Accepted, confident

The baseline agent, which stores facts without source labels and infers whenever any evidence is available, accepted all five. The pramāṇa-constrained agent identified problems in three.

The third scenario is the substantive result. A general principle held in the knowledge base (“this substance supports digestion”) is true in aggregate and inverts for a specific constitutional type with a specific presenting condition. The baseline applies the rule and produces a recommendation that would worsen the presenting symptom. The constrained agent identifies the reason as straying and refuses. No amount of attribution analysis on the baseline would have prevented this, because the baseline’s reliance on the rule was, from an attribution standpoint, entirely faithful. The problem was not opacity. The problem was that nothing was checking whether the reason licensed the conclusion.

This is a small, hand-constructed demonstration. It is not an empirical benchmark, and Section 7 treats that deficiency directly.

7. Honest Limitations and the Research Agenda

A positioning paper that lists only strengths is a marketing document. The following are the genuine weaknesses, stated as a reviewer would state them, each paired with the research programme that would address it.

7.1 No formal guarantees

The problem. SHAP’s credibility rests substantially on its axiomatic derivation. Pañcāvayava has no comparable formal characterisation. Without one, the framework is a well-motivated design pattern rather than a proven method, and it will be treated as such by formal methods reviewers.

The programme. Formalise pañcāvayava as a logic programme with stated axioms, and characterise the properties of the pramāṇa gate (soundness conditions, completeness limits, behaviour under source conflict). Establish what the ordering guarantees and what it does not. This is the single highest-value open problem in the agenda and is independently publishable.

7.2 The vyāpti establishment problem

The problem. Pañcāvayava’s third member requires a vyāpti, an invariable concomitance between reason and conclusion. Classical Nyāya devotes enormous attention to how vyāpti is established and how it survives the search for counter-instances (vyabhicāra). A computational system that simply asserts its general rules has helped itself to the hardest part of the theory. This is the deepest philosophical vulnerability in the approach, and a Nyāya scholar will identify it immediately.

The programme. Investigate computational procedures for vyāpti establishment and, more realistically in the near term, for *vyāpti challenge*: systematic search for counter-instances to a proposed general rule within available data. This connects naturally to the savyabhicāra detector, which is in effect a counter-instance check performed at inference time.

7.3 Representational flatness relative to Navya Nyāya

The problem. A tag-based representation of properties and conditions is substantially cruder than the Navya Nyāya analytic apparatus, which decomposes cognition into qualifier (prakāra), qualificand (viśeṣya), and relation (saṃsarga), permitting nested and relational specification of considerable precision. Using flat tags while invoking Nyāya is a real mismatch between the philosophy claimed and the philosophy implemented.

The programme. Design a representation informed by the Navya Nyāya triadic structure and evaluate whether the additional expressiveness yields detectable improvements in defect detection. This is genuinely novel work at the intersection of Sanskrit philology and knowledge representation, and it is the kind of contribution that would interest both communities.

7.4 No empirical validation

The problem. The claim that demonstrative traces are more trustworthy, more actionable, and more contestable than attribution plots is currently an assertion. No user study supports it. This is the objection a FAccT reviewer will raise first, and it is fair.

The programme. Controlled user studies comparing pañcāvayava traces against SHAP visualisations across three measured dimensions: *comprehension* (does the user correctly state why the decision was made), *calibration* (does the explanation help the user identify when the system is wrong), and *contestability* (can the user identify a specific step to challenge). This work requires institutional collaboration and ethics approval, and it is the necessary bridge from position paper to empirical contribution.

7.5 Ontology dependency and scaling cost

The problem. The udāharāṇa (exemplification) and śabda (testimony) components require curated domain knowledge. This does not transfer across domains without substantial expert effort, which

limits generality. Separately, per-decision generative construction of reasoning chains is expensive at production scale.

The programme. Pre-built ontologies for target domains with an interface for custom ingestion; template-driven trace construction with caching for recurring reasoning patterns, reserving generative construction for genuinely novel cases.

7.6 Responsible AI coverage gaps

The problem. The framework addresses transparency, contestability, and epistemic accountability. It does *not* address fairness, privacy, or lifecycle governance. A system can produce an impeccable demonstrative trace for a decision that is discriminatory, since the trace records the reasoning, not its distributive consequences. Claiming this as a complete responsible AI solution would be overreach.

The programme. Two directions. First, investigate whether Nyāya’s disputation theory supplies a native fairness concept. The distinction between *vāda* (truth-seeking debate, conducted under shared standards of evidence), *jalpa* (victory-seeking debate), and *vitaṇḍā* (purely destructive refutation) formalises something about *good-faith epistemic engagement* that has no obvious counterpart in statistical parity definitions. Whether this can be made precise and computational is an open and interesting question. Second, extend *hetvābhāsa* detection from logical defects to *ethical* defects: reasons that are formally valid but rest on impermissible grounds.

7.7 Gameability of the explanation

The problem. Any explanation format sophisticated enough to be persuasive is sophisticated enough to be manipulated. A well-formed *pañcāvayava* trace could be constructed to launder a decision made on other grounds.

The programme. Attribution-based faithfulness checking (Section 5.1), plus adversarial evaluation: can a trace be constructed that passes the gate and the defect scan while misrepresenting the actual basis of decision? Treating this as a red-team problem rather than an afterthought is the responsible posture.

8. Why This Matters Beyond the Technical Argument

8.1 A contribution rather than a branding exercise

There is a legitimate risk that framing this as “India’s XAI framework” invites dismissal as cultural positioning rather than technical contribution. The defence against that is not rhetorical. It is that the technical argument stands independently.

The claim is testable and specific: **a system that types its evidence by acquisition route and validates its reason-form before inferring produces explanations that are more faithful, more contestable, and more appropriately abstemious than one that does not.** That claim is either true or false regardless of the tradition it came from. The reason to name the tradition is not credit-seeking; it is that the tradition supplies the *integrated design*, worked out over two millennia, that would otherwise have to be invented from scratch and probably invented worse.

The correct framing is therefore: a contribution to global XAI research that happens to originate in an underexploited intellectual tradition, and which demonstrates that the tradition contains formally interesting computational content. Not: a national alternative to Western AI.

8.2 Policy and standards relevance

If the Article 86 correspondence in Section 3.4 survives formal analysis, it has direct implications. A demonstrative trace would constitute a candidate reference form for regulatory explanation requirements, which is a substantially more useful contribution to standards bodies than another attribution variant. Engagement pathways include NITI Aayog’s responsible AI work, BIS standards development on AI explainability, and the ISO/IEC 42001 implementation guidance ecosystem.

This should be pursued only after the formalisation in 7.1 exists. Approaching standards bodies with a design pattern is premature; approaching them with a formalised, empirically validated explanation form is not.

8.3 The domain that makes it non-optional

For AI systems operating over Indian knowledge systems, this is not one option among several. Ayurvedic reasoning is constitutionally individualised: the same intervention is indicated for one prakṛti (constitutional type) and contraindicated for another, and the tradition’s own logic for handling this is precisely the logic of when a general rule fails to apply to a particular case. That is savyabhicāra. A statistical recommender trained over population data cannot represent this, and an attribution method cannot explain it, because neither has the concept.

Building trustworthy AI over these domains requires an epistemic framework native to them. That is the applied case, and it is where the author’s own product work is directed.

9. Conclusion

The black box problem has been treated as a problem of *opacity*, to be solved by better instrumentation of the model’s internals. This paper argues it is more usefully treated as a problem of *warrant*, to be solved by requiring justification before permitting inference.

Nyāya Darshan supplies, in an integrated form, the three components this requires: a taxonomy of knowledge sources with a principled strength ordering, a taxonomy of reasoning defects, and a demonstrative output form built for public contestation. These were developed for human disputation and are, as it turns out, close to what automated decision systems now need and what regulation is beginning to demand.

The proposal is modest in its architectural claim and ambitious in its research programme. The architectural claim is a single inversion: justify, then decide. The research programme, set out in Section 7, spans formalisation, representation, empirical validation, and regulatory mapping, and is realistically a multi-year, multi-collaborator undertaking.

What has been established so far is a working demonstration that the inversion is implementable and that it changes outcomes in cases where source-blindness causes harm. What remains is the

harder and more valuable work of proving that it generalises, formalising what it guarantees, and measuring whether humans actually reason better with it.

The tradition has been available for two thousand years. The interesting question is what it can do now.

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